

MTH210 Project: Statistical Computing

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1. Data Description

The dataset used in this analysis was provided by the instructor.

- File number: fort.91
- Sample size: $n = 36$

2. Initial Values

The initial values required for the NR algorithm were computed using a moment-based approximation. The obtained starting values are:

$$[\alpha^{(0)} = 1.279167, \quad \lambda^{(0)} = 0.529513.]$$

3. Newton–Raphson Method

Since the likelihood equations cannot be solved analytically, the Newton–Raphson method was applied to obtain numerical solutions.

Stopping Criterion

The iterative process was stopped when: $[\max |\theta^{(k+1)} - \theta^{(k)}| < 10^{-6}.]$

Convergence

The algorithm reached convergence after **6 iterations**.

Maximum Likelihood Estimates

The estimated parameter values are: $[\hat{\alpha} = 2.386285, \quad \hat{\lambda} = 0.867887.]$

4. Bootstrap Methods

To evaluate the variability of the estimators, two bootstrap approaches were implemented.

Parametric Bootstrap

Random samples were generated from the fitted distribution using the inverse transformation method

Non-Parametric Bootstrap

Bootstrap samples were created by resampling the original data with replacement.

Details

- Number of bootstrap replications: $B = 500$

5. Results

Parametric Bootstrap Confidence Intervals

$$\alpha \in (1.559331, ; 4.313063)$$

$$\lambda \in (0.6324351, ; 1.279892)$$

Non-Parametric Bootstrap Confidence Intervals

$$\alpha \in (1.710158, ; 4.085206)$$

$$\lambda \in (0.6800646, ; 1.198798)$$